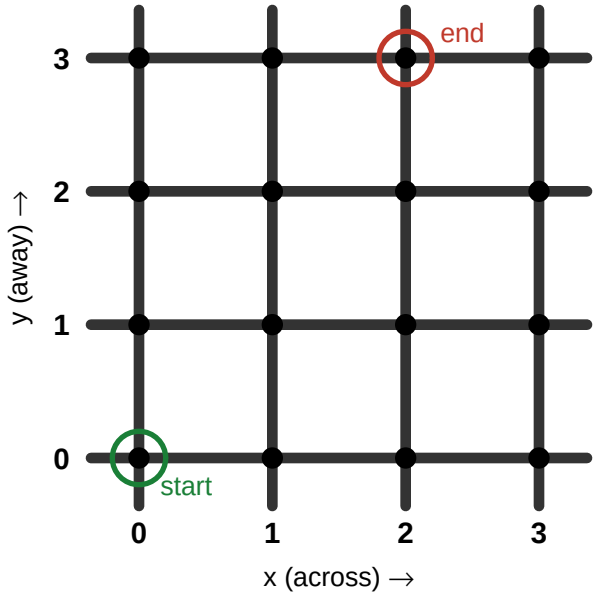
Lesson 4-4 · Trace it on paper

Name: _____ Date: _____

Trace (0, 0) → (2, 3) by hand: move right first, then up. The robot starts at (0, 0), so the path lists only the spots it moves to. Draw the path on the grid, then fill in the table.



Tips

- Start is (0, 0), circled green.
- End is (2, 3), circled red.
- Each move changes x or y by 1.
- Do all the moves across (x) first, then the moves up (y).
- Use one color for moves across, another for moves up.

Step	Action	New position	Path so far
1	x +1	(1, 0)	`[(1,0)]`
2			
3			
4			
5			

Row 1 is done for you to show the shape of it.

Questions

1. How many steps did it take? Does it match the Manhattan distance |2–0| + |3–0|?

   Steps: _____ Manhattan distance: _____ Match? Yes / No

2. Trace (1, 1) → (1, 1), the same spot. What is the path, and how many steps?

   Path: _____ Steps: _____

3. Trace (2, 0) → (2, 3). Write the full path. (Remember: the start is not included.)

   Path: _____ Steps: _____